

Implied-Volatility Dynamics, Dealer Positioning and Intraday Reversals in QQQ

An event-conditioned and forward study of zero-day-to-expiry options, 2022–2025

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Working paper · 17 September 2026 · Descriptive and pre-registered tests; no trading rule was fitted

ABSTRACT. We ask what the zero-day (oDTE) option market looked like at intraday reversals in QQQ, and whether any of it predicts whether a price level holds. Six studies are run on three sources: 665 days of one-minute oDTE chains with open interest (2022–2024), 224 days of one-minute OPRA quotes for every expiry (2025), and a year of twice-daily full-chain snapshots (2022), all graded against one-minute Nasdaq-100 futures bars converted to QQQ terms. A rule harness that ports a five-pass positioning-alignment methodology produces hold rates at the random-walk base rate in every class, in-sample and in a 2024 holdout. An event atlas of 1,681 realised turns and 371 run-throughs finds positioning at the strike indistinguishable between the two, while the direction of at-the-money implied volatility over the prior 30 minutes separates them (rising into 53% of bottoms versus 17% of downside run-throughs), a contrast that replicates on unseen 2025 data (58% versus 25%). As a pre-registered rule on every strike approach the same state is a modest screen (35.6% versus 30.7% on a 1:2 bracket), with one strong negative: a put-side level reached on falling implied volatility held 21–24% across four datasets. Whole-chain walls, the strike in front of a wall, breakout continuation, retest fades and overnight brackets are each tested and found at or below base rate. The first-hour direction persists to the close (73%). We conclude that positioning locates where price interacts with hedging and the vol state decides the interaction, and that the economically usable content is a filter on what to skip rather than a signal on what to take.

1. Introduction

A large practitioner literature reads dealer option positioning — gamma, vega, charm and vanna exposure by strike — as a map of where intraday price will reverse. The claim is rarely tested against the realised price path at the resolution the trade is placed, and when it is, the test is usually a strategy backtest whose parameters were chosen on the same data. This paper takes the opposite route. Events are selected from the realised path, the option market at each event is described, descriptions are compared across event classes, and only then are the surviving descriptions turned into pre-registered rules and tested on data not yet examined.

Three questions are addressed. First, does positioning at a strike, however measured, separate the strikes that held from the strikes that were run through? Second, does anything in the option tape into the level separate them? Third, once a level breaks, what does the option book say about continuation, retest and target? A fourth question, whether the same construction works overnight, is answered on the one dataset that permits it.

The design rules are fixed throughout. Every barrier scales to a contemporaneous volatility measure or is the trader's own fixed bracket. No threshold was chosen by reference to an outcome. Every result is split by year before being read pooled, and each test carries a null (a symmetric bracket, a displaced placebo, or a random-walk base rate) that is read before any table.

2. Data

2.1 Zero-day chains, 2022–2024

For each of 672 trading days (3 January 2022 to 31 December 2024; 665 usable), one record per minute, strike and side for every QQQ option expiring that day, 09:31 to 15:59 ET: bid, ask, mid, the pre-open open interest, a single per-strike implied volatility shared by call and put, the parity forward F and time to expiry T in years measured as minutes to the 16:00 close over a 365-day year. The file's delta, gamma and per-strike gamma exposure were verified to reproduce the forward model of Section 3.1 with zero rate to displayed precision. Coverage grew with the product: about 130 strikes per minute in 2022, 165 in mid-2024, 326 on the December 2024 expiry. On some days every strike carries a constant fractional offset near 0.22 (a distribution adjustment at the source); these were restored to whole strikes.

2.2 OPRA quotes, 2025

224 trading days (10 February to 30 December 2025) of one-minute consolidated best bid and offer for every listed QQQ option contract (all expiries; roughly 8,700 contracts and 3.4 million rows per day, 08:01 to 16:16 ET). No open interest and no implied volatility are supplied. The forward is recovered from put–call parity on near-the-money pairs and implied volatility is solved from the out-of-the-money mid (Section 3.1). This set was not opened until the 2022–2024 work was complete, and each of its tests was run once.

2.3 Full-chain snapshots, 2022

Two JSON snapshots per calendar day for 2022 (09:36 and 15:51 ET) of the entire QQQ chain: every expiry (about 32) and strike, with open interest, implied volatility, delta and gamma per contract, from a retail brokerage feed (15-minute delayed quotes; open interest is a daily figure and unaffected). 363 days carry both snapshots; 167 join to the 2022 minute chains above.

2.4 Futures bars and conversion

One-minute open, high, low, close of the front-month Nasdaq-100 futures contract in Eastern Time, 2010–2026. Bars are expressed in QQQ terms by dividing by a rolling ratio,

$\rho_t = \text{median}_{t-30} u$

lagged one minute so that no bar is converted with a ratio that uses its own close. Residual basis noise is about 0.1–0.2 index points, comparable to the fill tolerance and therefore a floor on the precision of any "exact strike" statement. Overnight tests use the 15:59 ratio for the whole night.

3. Model, exposures and definitions

3.1 Forward model

All greeks use the Black (1976) forward model with zero rate, the model the 2022–2024 file was generated with. With F the forward, K the strike, σ the per-strike implied volatility and T the time to expiry in years,

$$d_1 = \frac{\ln(F/K) + \frac{1}{2}\sigma^2 T}{\sigma\sqrt{T}}, \quad d_2 = d_1 - \sigma\sqrt{T},$$
$$C = F N(d_1) - K N(d_2), \quad P = K N(-d_2) - F N(-d_1),$$

$$\Delta_C = N(d_1), \quad \Delta_P = N(d_1) - 1, \quad \Gamma = \frac{\varphi(d_1)}{F\sigma\sqrt{T}}, \quad \mathcal{V} = F\varphi(d_1)\sqrt{T},$$

$$\text{charm} = \frac{\partial \Delta}{\partial t} = \frac{\varphi(d_1) d_2}{2T}, \quad \text{vanna} = \frac{\partial \Delta}{\partial \sigma} = -\frac{\varphi(d_1) d_2}{\sigma},$$

where t is calendar time (so charm is $-\partial \Delta / \partial T$), N and φ are the standard normal distribution and density, and charm and vanna are identical for the call and the put at a strike. For ODTE contracts $T = m/525,600$ with m the minutes remaining to 16:00 ET, floored at 30 minutes. For the 2025 quotes, σ is obtained by bisection on the out-of-the-money mid (put below F , call above), 40 iterations on $[0.01, 6]$, discarded where the mid is at or below intrinsic value.

3.2 Dealer exposures and hedging-flow signs

Under the standard convention that dealers are long the calls and short the puts held by customers, the per-strike exposures summed over sides $s \in \{C, P\}$ with sign $\epsilon_C = +1$, $\epsilon_P = -1$ are

$$\text{GEX}_K = \sum_s \epsilon_s \Gamma_s \text{OI}_s \cdot 100 \cdot S^2 \cdot 0.01, \quad \text{VEX}_K = \sum_s \epsilon_s \mathcal{V}_s \text{OI}_s \cdot 100, \quad \text{DEX}_K = \sum_s \epsilon_s \Delta_s \text{OI}_s \cdot 10$$

Two second-order quantities are expressed as hedging *flows* so that their sign has one reading: positive means dealers must buy the underlying. As time passes the dealer's delta changes by $\sum_s \epsilon_s \text{OI}_s \cdot 100 \cdot \text{charm}$ and the hedge must move the opposite way; as implied volatility falls by $d\sigma$ the hedge must move by $+\sum_s \epsilon_s \text{OI}_s \cdot 100 \cdot \text{vanna} d\sigma$. Hence

$$\text{charm flow}_K = -\sum_s \epsilon_s \text{OI}_s \cdot 100 \cdot \text{charm}, \quad \text{vanna flow}_K = +\sum_s \epsilon_s \text{OI}_s \cdot 100 \cdot \text{vanna},$$

with the second read as the flow when implied volatility *falls*. Magnitudes are compared only within a greek, never across greeks, by ranking or by ratio to the band median. On zero-day contracts **VEX** and **GEX** are the same map, since both are proportional to $\varphi(d_1)$ when T is hours.

3.3 Expected move and the implied-volatility state

The expected move to the close at minute t is $E_t = S_t \sigma_t^{\text{ATM}} \sqrt{T_t}$, with σ^{ATM} the implied volatility of the strike nearest F (2022–2024) or the inverse-distance weighted mean of strikes within one point of F (2025). The state of the tape at minute t is the 30-minute change $\Delta_{30} = \sigma_t^{\text{ATM}} - \sigma_{t-30}^{\text{ATM}}$, classed *rising* if $\Delta_{30} > 0.01$, *falling* if $\Delta_{30} < -0.01$, else *flat*; and the retreat from the trailing 60-minute peak, $\max_{t-60 \leq u \leq t} \sigma_u^{\text{ATM}} - \sigma_t^{\text{ATM}}$, called *rolled over* when it exceeds 0.01. The window and threshold were fixed before any outcome was computed and were not varied.

3.4 Implied-volatility walls

The inner walls are the strikes at which $|\Delta|$ equals $\delta^* = 0.1925$ along each out-of-the-money wing of the front-expiry smile, using the model above with $r = 0.04$ in d_1 as the specification requires. Points that violate monotone decreasing $|\Delta|$ outward from spot are dropped, and the crossing is linearly interpolated between the bracketing strikes a, b :

$$K^* = K_a + \frac{|\Delta_a| - \delta^*}{|\Delta_a| - |\Delta_b|} (K_b - K_a).$$

The outer walls sit at fixed fractions of spot beyond the inner ones, $K_u^{\text{out}} = K_u + S \cdot 1.56/750$ and $K_l^{\text{out}} = K_l - S \cdot 1.79/750$, the lower band wider for put skew. A bracket is *frozen* when computed

once (at the open, at 10:00, or from the prior evening's chain) and *live* when recomputed every minute.

3.5 Events

Turns. A zigzag on the converted highs and lows with reversal threshold g (the target below): a local extreme is a turn when the legs into and out of it both reach g .

Run-throughs. One of the three largest $|GEX|$ strikes within 1% of spot at 10:00, crossed after 10:00 with continuation of at least g beyond it and no turn within 0.5 of it for the rest of the session.

Approaches. The first minute a side of price comes within 0.5 of a whole strike within 1% of spot (support: previous close above $K + 0.5$ and low at or below it; resistance mirrored), one per (day, strike, side); the state is read one minute before contact.

Breaks. The first one-minute close at or beyond $K \pm s$ after a close on the other side of K that day, with s the stop below. **Retest.** The first return to within 0.5 of a broken strike from the far side.

3.6 Outcome grading

Every level is graded as a resting limit at the strike with the trader's fixed bracket, stop $s = 40/40.7 = 0.983$ and target $g = 80/40.7 = 1.966$ QQQ points (40 and 80 points of the Nasdaq-100 micro contract at the prevailing ratio), fill tolerance 0.15. A limit fills only when price trades *through* it ($\$L_j$ held, records whether the level rejected by g before overshooting by s from contact. Continuation trades after a break fill at the break bar's close. Expectancy per fill is $\hat{p}g - (1 - \hat{p})s$ plus the marked value of flat closes, reported in micro-contract points.

3.7 Nulls and intervals

For a driftless walk with absorbing barriers at $-s$ and $+g$ the probability of reaching the target first is $s/(s + g)$ (the gambler's-ruin result), so the base rate of the 1:2 bracket is $1/3$ and of a symmetric bracket $1/2$. Both are checked directly: a symmetric bracket at arbitrary strikes returned 48.3% and the 1:2 population 31–33%. Hold rates carry 95% Wilson intervals,

$$\hat{p}_{\pm} = \frac{\hat{p} + \frac{z^2}{2n} \pm z \sqrt{\frac{\hat{p}(1-\hat{p})}{n} + \frac{z^2}{4n^2}}}{1 + \frac{z^2}{n}}, \quad z = 1.96.$$

Displaced placebos move a level by $0.25E$ while keeping its labels; a level-specific effect should vanish under displacement.

4. Study I — a positioning-alignment rule harness

A five-pass methodology in use for scoring strikes (structural role from concentration relative to the band; forced-flow alignment across gamma, charm, vanna and delta with a fixed truth table; tenor durability; live evidence; regime coherence, mapped to ordinal probability tiers) was ported line for line and run on every strike within 1% of spot at 15-minute decision ticks from 09:45 to 15:30 over 2022–2024, 11,031 first-appearance calls. Charm and vanna came from Section 3.1 on the file's own inputs. The 2024 year was held out and read once.

Cut (win rate given a resolved fill, bracket A)	2022	2023	2024 holdout
Population, all strikes	32.8%	30.7%	30.4%
Alignment vector "aligned"	27.7%	35.5%	30.9%
Alignment vector "opposed"	32.2%	30.4%	27.6%
Gamma, vega and charm all agreeing	25.4%	36.2%	32.4%
The day's major wall	27.7%	29.8%	31.7%
Highest-tier probability class (62)	25.0% (in-sample)		29.0%

Table 1. The harness reproduces the base rate in every class. An era-normalised bracket (scaled by spot/700) gave the same picture. Displaced placebos matched the real strikes. Higher gamma concentration was associated with slightly lower hold rates (34% at the lightest strikes to 29% at the heaviest).

Two effects that are not greeks did register: strikes more than half an expected move away fill rarely and, when they fill, hold 25%; and the hold rate decays from 33% at 09:45 to 18% at 15:30, partly mechanical since a target twice the stop is seldom reached in the last half hour.

5. Study II — the reversal atlas

Rather than ask whether a rule pays, this study asks what the option market looked like at the moments price actually turned. From the 2022–2024 minute chains and bars, 1,681 turns (862 bottoms, 819 tops) and 371 run-throughs were extracted, and for each the ladder of eleven strikes centred on the event was reconstructed one minute earlier together with a 60-minute tape of the event strike. A stratified sample of 102 events was read by hand to form hypotheses; each was then computed over the population and split by year.

5.1 Positioning locates the interaction and does not decide it

Class	n	Local OI wall within 1 pt	Event strike is ± 3 max of GEX	Nearest wall
Turn, bottom	862	38%	21%	one strike below
Turn, top	819	34%	19%	one strike above
Run-through, down	179	39%	44% (selected)	one strike below
Run-through, up	192	38%	44% (selected)	one strike above

Table 2. A local wall is a strike whose defending-side OI is at least twice the ladder median; chance for the GEX column is about 14%. Turns did not cluster on whole strikes (22% within 0.10, uniform 20%; 5% within 0.15 of a multiple of five, uniform 6%).

Charm flow was positive at 63–69% of bottoms and at 65–80% of downside run-throughs; its sign is set by the strike's position relative to spot, not by the outcome. Vanna flow at the event strike was near zero in every class, as it must be at the money.

5.2 The direction of implied volatility into the event

Class	n	ATM IV rose > 1 pt	ATM IV fell > 1 pt	Median change
Turn, bottom	862	53%	23%	+1.16
Run-through, down	179	17%	53%	–1.18
Turn, top	819	27%	51%	–1.07
Run-through, up	192	7%	67%	–2.05

Table 3. Change in ATM oDTE implied volatility over the 30 minutes before the event, in volatility points. The bottom versus downside run-through contrast held in each year (54/17, 41/15, 57/19) and replicated on 2025 (58% rising into bottoms, 25% falling; tops 33/48).

Two refinements. At 45% of bottoms implied volatility had already retreated more than one point from its 60-minute peak before the price low printed. And into a real low the downside wing (strikes two to five points below) rose *less* than the at-the-money strike (median wing-minus-ATM change –1.45; 57% decompressing by more than a point) whereas into a run-through the wing stayed relatively bid (–0.48; 40%). The repricing at a real low is in the front strikes, not the tail.

5.3 Where the turn prints relative to the wall

Among turns with a local wall within three points beyond them, 17% printed within 0.35 of the wall, 37% between 0.35 and 1.5 in front, and 46% between 1.5 and 3 in front. Nothing tested — wall dominance, whether it was the day's largest, time of day, approach speed, OI on the front strike — separated the exact cases from the rest. Using a practical fill window (0.12 above to 0.37 through a strike), 49% of lows fell in some strike's window, exactly the uniform expectation, and the strike that caught a turn was the local maximum of any ladder feature among its four neighbours at chance rates (17–22% against 20%).

6. Study III — the state as a rule: forward test and heavy strikes

The description of Section 5.2 was turned into a pre-registered rule and run on the 2025 quotes: at every first approach to a whole strike within 1% of spot, classify Δ_{30} one minute before contact and grade the bracket. 2,886 approaches on 217 days.

IV state at approach (2025)	Approaches	Win filled	95% CI	Level held	Expectancy / fill (MNQ pts)
Rising	1,041	35.6%	32–39	52%	+3.4
Flat	698	32.8%	29–37	51%	+1.1
Falling	1,147	30.7%	28–34	51%	–1.9

Table 4. Break-even on the 1:2 bracket is 33.3%. The direction held in 10 of 11 months; the resistance side was cleaner (39.9% rising versus 29.0% falling). Approaches on rising volatility that had already rolled over won 32.5% against 37.0% for those still rising: vol exhaustion describes turns but does not time entries.

The same test on 2022–2024, which carries open interest, shows where the state bites. Heavy strikes (defending-side OI at least twice the band median, or among the three largest |GEX| at 10:00)

approached from above on falling volatility won 23.8% (-10.3 points per fill, $n = 187$) against 33.1% on rising; light strikes (OI below half the median) won 34–38% regardless of state. Heavy strikes as a class held less often than light ones in every state.

7. Study IV — implied-volatility walls

Brackets of Section 3.4 were built every minute on 665 days. The live bracket's inner width falls from 6.6 points at the open to 1.6 at the close, a $\sqrt{T}$ envelope; a wall moves a median 0.3–0.4 points per 15 minutes and sweeps 1.3–1.5 E_0 over the day. Frozen at the open, the bracket's inner half-width is 0.85–0.91 E_0 .

Frozen-at-open wall	Days reached	Win filled	95% CI	Held	Placebo (+0.25 E) win
Lower inner	37%	34.3%	28–42	50.5%	31.4%
Lower outer	25%	28.7%	21–38	43.5%	26.0%
Upper inner	41%	28.1%	22–35	48.9%	22.5%
Upper outer	28%	28.9%	21–38	44.9%	27.3%

Table 5. The live bracket, being a moving target, filled at roughly half the frozen rate and did worse. Lower walls reached on falling implied volatility won 21% (-10 to -15 points per fill); on rising, 34–36%.

Overnight (2025, 169 nights), the prior-evening bracket built from the next expiry at 15:59 with $T = 24$ h sits 0.83 E from spot and was reached by an inner wall on 7% of nights and an outer wall on 3–4%, too few events to grade. Re-anchoring T to 04:00 (10 h) pulls the walls to 0.55–0.59 E and raises reach to 20–22% of nights, at which the walls held 30–33% on about twenty events; whole-strike contacts overnight won 29.9% on 686 events. The overnight session travels a fraction of a day's expected move, and nothing in the bracket construction changes that.

8. Study V — breaks, retests, targets and day bias

4,149 breaks (Section 3.5) within 2% of the 10:00 print on 660 days, 2022–2024, with 1,912 more on 2025 without open interest.

Continuation entered at the break close, 40/80 bracket	n	Win	Expectancy / fill
Heavy strike, all	1,873	32.9%	+1.7
Light strike, all	2,276	31.9%	+0.5
Heavy, broken down on falling IV	314	37.8%	+6.2
Heavy, broken up on rising IV	157	24.1%	-4.9
Any strike, broken up on rising IV (2025)	221	25.6%	-5.4
Heavy, positive-gamma day	936	37.0%	+6.0
Heavy, negative-gamma day	937	28.9%	-2.6

Table 6. Continuation at the break is break-even; the squeeze breakout (up on rising volatility) fails in both datasets.

Retests. 75% of breaks retest the broken strike, median 12–13 minutes later (80–88% and 6–10 minutes on 2025). A reversal placed at the strike from its new side — the classic break-and-retest continuation — won 28.7% at heavy strikes and 34.3% at light ones; the implied-volatility state at the retest did not discriminate (25–30% in each class). At heavy strikes, that is, the retest more often reclaims the level than confirms the break.

Targets. A break runs a median $0.8E$ beyond the level before price returns to it. The next heavy strike was reached before a return 96% of the time when within $0.4E$, 71% at $0.4-0.8E$, 42% at $0.8-1.5E$ and 13% beyond. A trade stopped through the level and targeting the next wall won 71% of the time and lost money, since the target is usually smaller than the stop.

Day bias. The direction of the first hour (09:31 to 10:30) matched the direction of the close on 73% of 374 days (78%, 68%, 75% by year). Measured strictly, the rest-of-day move in the first-hour direction had median $+0.10$, $+0.18$, $+0.30$ and $+0.01E_0$ for first-hour moves of under 0.25 , $0.25-0.5$, $0.5-0.8$ and over $0.8E_0$ (persistence 62%, 75%, 83%, 86%). The first turn of the 10:00 hour was followed by a close in the opposite direction 67–70% of the time; it is the counter-trend bounce inside the open drive. Realised range was $1.29-1.39E_0$ by gamma regime; the close-to-open move $0.65E_0$.

9. Study VI — the whole chain

The objection to Sections 4–8 is that zero-day positioning is thin and two-sided, and that the walls practitioners trade are whole-book walls. The 2022 full-chain snapshots allow the test on one year. Per strike, **GEX** was summed across all expiries from the 09:36 chain (and from the prior evening's 15:51 chain as a second variant), and the approach test of Section 6 was rerun on the 167 days that join to the minute chains.

Strike class by the 09:36 whole-chain book (2022, 2,155 approaches)	n	Win filled	95% CI	Expectancy / fill
Heavy on the whole chain and on oDTE	372	31.7%	27–37	−0.8
Heavy on the whole chain only	361	31.9%	27–38	−0.9
Heavy on oDTE only	354	33.9%	29–40	+1.7
Light	1,068	34.3%	31–38	+2.1
The named call/put gamma or OI wall	162	28.9%	22–37	−4.4
Defending OI above 10× the band median	231	25.4%	20–32	−7.7
Defending OI below 0.5× the band median	181	42.6%	35–51	+11.7

Table 7. Whole-chain heavy support reached on falling implied volatility won 24.3% (−10.8 per fill, $n = 137$); on rising, 35.2%. The prior-evening book gave the same ordering (heavy 30.6%, light 34.8%).

A natural reading of the empty-strike result together with Section 5.3 is that the tradeable level is the light strike directly in front of a wall. Tested on both books, it is not: 30.0% on the whole-chain book (against 35.3% for light strikes with no wall within two) and 32.0% on 2022–2024 zero-day (against 32.4%). A 39.9% cell for "wall two strikes behind" on the whole chain is absent on the larger zero-day set (31.8%) and is treated as noise.

10. Synthesis

The six studies agree on a division of labour. Positioning — open interest, gamma, vega, charm and vanna, on the day's expiry or on the whole book — locates where price interacts with dealer hedging: those strikes fill more often (84–91% against 77–80%) and price is drawn to them, with late-session concentrations pulling price toward them and the extreme printing one to four points short. Positioning does not decide the interaction: the heaviest strikes held least, the named walls held less than the population, and the strike that catches a turn is a coin flip among its neighbours on every ladder feature.

What decides it is whether there is a bid for protection to exhaust. A low that holds is reached with implied volatility rising, the repricing concentrated in the front strikes, and the volatility bid often exhausting before the price low; a level that is run through is reached on falling volatility, orderly, with the wing still bid. The upside mirrors this with one addition: a rally on falling volatility carries the vanna tailwind and grinds through call-heavy strikes, while a rally on rising volatility is a squeeze and its breakouts fail.

Translated into decisions on a fixed 1:2 bracket, the content is modest and mostly negative. The volatility state is a screen worth about five points of win rate on the population, ten at put-side walls, with one rule strong enough to stand alone: a put-side level reached on falling implied volatility held 21–24% in four independent datasets and should not be faded. Continuation at a break, break-and-retest at the strike, resting at the wall, resting one strike in front of it, and the overnight bracket each test at or below the base rate. Targets scale in expected moves, with the next heavy strike reachable inside $0.8E$ and unlikely beyond $1.5E$. The first hour carries a real but modest directional bias into the close. Whatever precision a trader has in *which* strike turns comes from the price action at the level, not from the option book.

11. Limitations and the question of overfitting

No parameter was fitted. The bracket, the fill tolerance, the 30-minute window, the one-point cutoff, the two-times-median wall definition, the 60-minute tape span, the distance bins and the first-hour bins were each set once, in the scripts, before the corresponding statistics were computed, and none was varied to improve a result. There is no model, no coefficient and no rule whose performance was maximised.

The weaker risk is in-sample description. The hypotheses of Section 5 were formed by reading 102 events drawn from the same 2,052 used to check them. Three things bound the damage: the population is twenty times the sample; every main contrast was split by year and held in three regimes; and the features that failed the check are reported beside those that passed. The decisive discipline was Section 6: the same description, as a rule fixed in advance, on a year of quotes that had not been opened, read once. It replicated as a description (58/25) and shrank as a rule (35.6/30.7), which is what an honest forward test of a real but small effect should look like.

Selection on outcome. Sections 5 and 9 condition on realised events; Sections 6–8 do not.

Class asymmetry. Run-throughs were drawn from heavy strikes crossed after 10:00; turns from the whole session. The time-of-day contrast is partly definitional and the positioning comparison is biased toward finding positioning at run-throughs, against the conclusion.

Confounding. Implied volatility rises when price falls. The statistic in Table 3 is informative only in contrast with the run-throughs, on which price was also falling.

Data. Per-strike implied volatility is parity-consistent, so no put–call skew within a strike; open interest is a daily snapshot; trade volume is absent everywhere; the futures conversion carries 0.1–0.2 points of basis noise; the whole-chain test is one year, a bear market with volatility spikes, on delayed quotes; the overnight test has too few events to say more than that the bracket is rarely reached.

12. Reproducibility

Every study is one script over the named sources with its thresholds in its header; the descriptive statistics are computed by short report scripts; the stratified sample is seeded. The 2024 holdout and the 2025 quotes have each been read once and should not be re-cut. A further test requires data not on hand: a second year of full-chain snapshots to confirm Section 9 outside 2022, or trade prints to restore the live-evidence pass of the harness.

Working paper, 17 September 2026. Sources: one-minute QQQ oDTE chains with open interest (2022–2024); one-minute OPRA consolidated quotes for all QQQ expiries (2025); twice-daily full-chain QQQ snapshots (2022); one-minute Nasdaq-100 futures bars. Descriptive and pre-registered; no trading rule was fitted or optimised.